SYSTEM AND METHOD FOR AN ENCRYPTED DATA BUS

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Title: SYSTEM AND METHOD FOR AN ENCRYPTED DATA BUS

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Abstract

A system and method for securing data transmission over a physical data bus. The system comprises a bus controller and a plurality of peripheral devices, wherein all data transmitted between the controller and the devices, including command and control signals, is encrypted. Each device is equipped with a cryptographic processing unit to encrypt and decrypt data in real-time. The system further includes a hardware-enforced mechanism within each peripheral device that governs whether it can accept communications from other peripheral devices. This mechanism is configured at the firmware level and cannot be overridden by commands from the bus controller, thereby preventing unauthorized inter-device communication and mitigating the risk of malicious firmware compromising the bus.

Background of the Invention

Conventional data bus architectures, such as the Universal Serial Bus (USB), typically treat the physical bus as a trusted channel. This trust is based on the legacy assumption that any

device connected to the bus is physically proximate to the host computer and is operated by a benevolent user. However, in modern disaggregated computing environments and with the proliferation of complex supply chains, this assumption is no longer valid. Malicious actors can compromise the firmware of peripheral devices (e.g., webcams, keyboards, network adapters) to eavesdrop on bus traffic or to initiate unauthorized communications with other peripherals, creating a significant security vulnerability.

Summary of the Invention

The present invention is a system and method for a comprehensively encrypted data bus that enforces cryptographic security and granular control over inter-device communication. The system ensures that all data traversing the bus is encrypted, rendering it unintelligible to unauthorized observers.

The system features a bus controller that establishes encrypted communication channels with each connected peripheral device. Each peripheral device contains a dedicated cryptographic unit to handle encryption and decryption, ensuring that even the device's own firmware cannot access plaintext data intended for other components. A key innovation is the inclusion of a firmware-level, hardware-enforced switch within each peripheral device. This switch controls whether the device is permitted to communicate directly with other devices on the bus. This setting is immutable via bus commands, meaning the bus controller cannot instruct a device to change its policy. This provides a robust defense against attacks where a compromised device is used as a vector to attack other peripherals.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram of the encrypted bus system, showing the encrypted communication channels between the bus controller and peripheral devices.

FIG. 2 is a conceptual diagram illustrating the hardware-enforced communication policy switch within a peripheral device.

Detailed Description of the Invention

Referring to **FIG. 1**, the encrypted bus system (100) is shown. A Bus Controller (110) is connected to a plurality of peripheral devices, including a Webcam (120) and a Network Card (130), via a physical bus (140). All data traffic between the controller and the devices (150) is encrypted. Each device possesses a cryptographic unit (121, 131) that manages the encryption and decryption of data.

Referring to **FIG. 2**, a peripheral device (200) is shown in detail. The device includes its main functional logic (210) and a cryptographic unit (220). A hardware-enforced policy switch (230) is implemented at the firmware level. In this example, the switch is set to "DENY" (240), which prevents the device from accepting any communication from other peripheral devices on the bus, even if commanded to do so by the Bus Controller. This policy is immutable via the bus interface, providing a high level of security against malicious commands.

Claims

What is claimed is:

1. A secure data bus system, comprising:
    - a bus controller;
    - a plurality of peripheral devices communicatively coupled to the bus controller; and
    - a cryptographic unit within each peripheral device,
    - wherein all data transmitted over the bus between the bus controller and the plurality of peripheral devices is encrypted.

2. The system of claim 1, wherein the encrypted data includes command and control signals.

3. The system of claim 1, further comprising a hardware-enforced policy mechanism within each peripheral device that determines whether the device can accept communications from other peripheral devices.

4. The system of claim 3, wherein the policy mechanism is configured at the firmware level of the peripheral device.

5. The system of claim 3, wherein the policy mechanism cannot be altered by commands received from the bus controller.

6. A method for secure communication on a data bus, comprising:

   - establishing, by a bus controller, an encrypted communication channel with a peripheral device;
   - transmitting encrypted data, including command and control signals, from the bus controller to the peripheral device;
   - decrypting the encrypted data at the peripheral device using a cryptographic unit; and
   - enforcing, at the peripheral device, a hardware-level policy that restricts communication with other peripheral devices on the bus.

Drawings

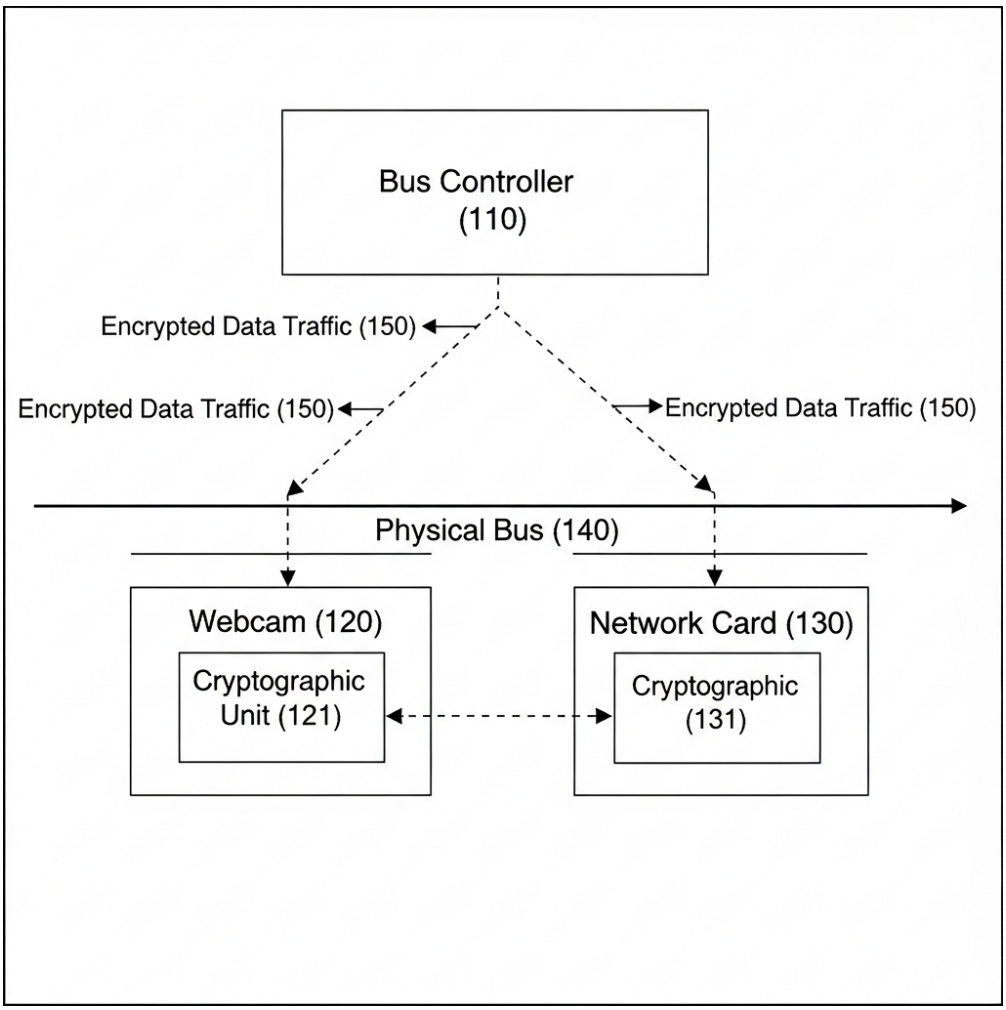


Figure 1: Block diagram of the encrypted bus system, showing the encrypted communication channels between the bus controller and peripheral devices.

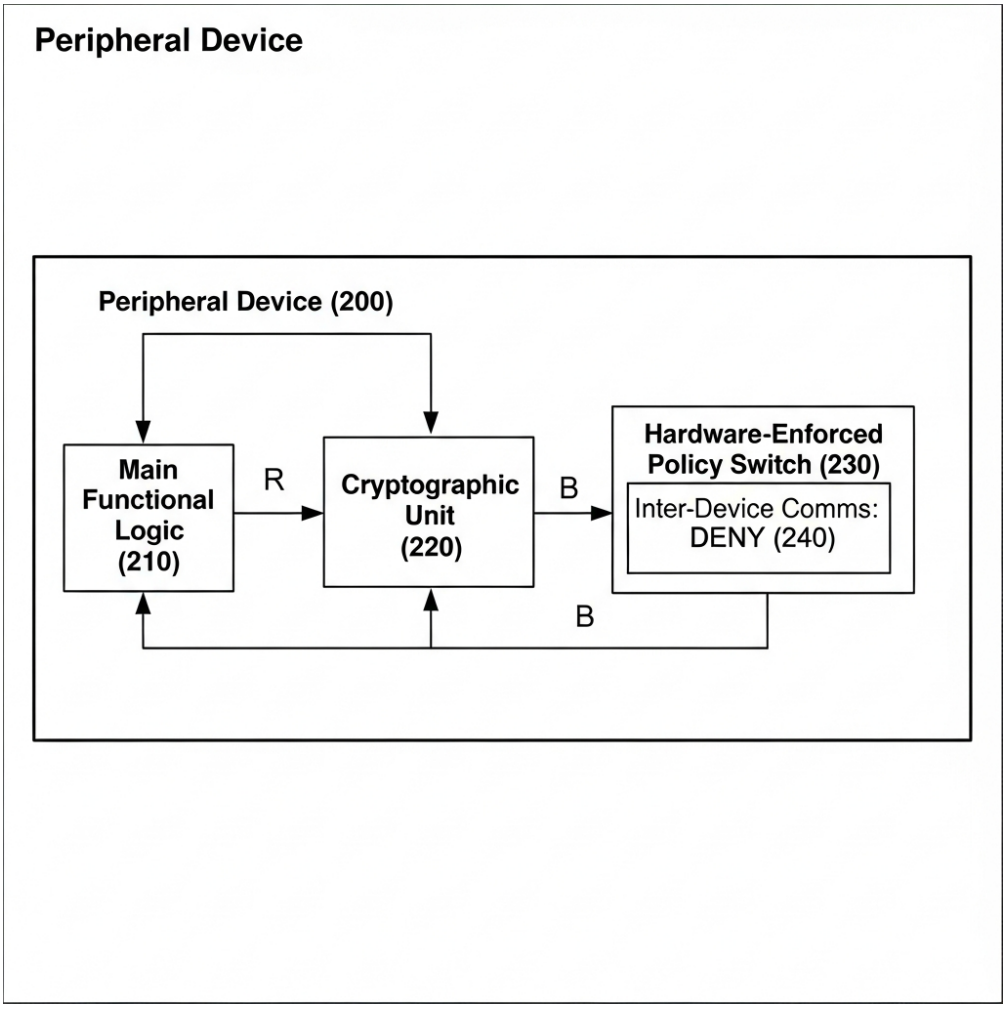


Figure 2: Conceptual diagram illustrating the hardware-enforced communication policy switch within a peripheral device.